ARCHIT GUPTA

Apex, NC 27502, 919-432-2583, 919-432-2583, architgupta0602@gmail.com

PROFESSIONAL SUMMARY

Results-driven data science intern with Qualitest Group who developed an NLP-powered skill-mapping application and engineered backend services. Proficient in Python and automation testing, I excel in collaborating with cross-functional teams, enhancing client relations, and delivering impactful AI-driven solutions that streamline workflows and optimize resource allocation.

SKILLS

- Agentic AI
- JIRA, Smartsheet, SpiraPlan
- Automation Testing
- Python
- Github Copilot, Claude Code, Codex
- Playwright Automation
- Agile
- Client Relations
- Custom MCP

WORK EXPERIENCE

Generative AI Architect — August 2026 - Current
QualityAI

- Leading the development of **QEAgents**, an agentic AI-powered quality engineering and test automation pipeline for Ellucian Banner and Colleague, achieving a **90% reduction in test automation costs** through the transition from traditional automation workflows to AI-driven agent orchestration.
- Architected a multi-agent AI pipeline comprising specialized agents for requirements analysis, test planning, automated test generation, self-healing test execution, and intelligent failure triage.
- Developed custom **Model Context Protocol (MCP) integrations and tooling**, connecting AI agents with Jira, Smartsheet, and SpiraTest through REST APIs to enable intelligent access to requirements, test management systems, and QA artifacts.
- Engineered Playwright-based **agentic automation** workflows to autonomously generate TypeScript test scripts, analyze execution failures, and support automated test healing and pull request generation.
- Designed a local-first agent execution architecture integrating **Claude Code CLI, AWS Bedrock, custom MCP services**, Playwright, GitHub, and Jenkins CI/CD workflows, incorporating human-in-the-loop validation checkpoints.

Data Science and Artificial Intelligence Intern — May 2022 - August 2026
Qualitest Group

- Architected an end-to-end **GenAI document extraction platform** to automate extraction of 16 structured NIAD fields from complex PDF and DOCX syndicate packages and term sheets using schema-guided LLM workflows.
- Designed a **multi-stage LLM architecture** incorporating document routing, section-level extraction, prompt engineering, deterministic validation, and business-rule enforcement to improve extraction accuracy and reliability.
- Implemented a **human-in-the-loop feedback and retrieval workflow** using PostgreSQL/pgvector embeddings, enabling reviewer corrections to be reused as few-shot examples and continuously improve future extractions without model retraining.
- Designed secure **REST API-based processing pipelines** with TLS/OAuth2, document de-duplication, in-memory document processing, structured JSON responses, and a no-document-retention architecture to support secure integration with downstream applications.
- Optimized GenAI inference performance through **parallel LLM execution, prompt caching, selective page routing, and model tiering**, reducing unnecessary token usage, latency, and per-document processing cost while reserving higher-capability models for ambiguous extraction tasks.

- Evaluated and benchmarked **OpenAI and Anthropic LLM models** across accuracy, latency, token consumption, and cost to define a scalable model-selection strategy for production document-processing workloads.

Customer Success Manager Intern December 2025 - May 2026
AccelQ
- Consulted with clients to understand web application functionality, assess testing requirements, and recommend tailored test plans and software quality services.
- Collaborated with client stakeholders and cross-functional teams, **translating technical requirements** into practical testing recommendations and communicating solution value.

Software Engineering Intern May 2025 - July 2025
Anomali Inc. | Redwood City, CA
- Spearheaded a malware attack-flow extraction feature mapping attack patterns, threat actors, and techniques to MITRE ATT&CK identifiers.
- Developed **Python threat-detection algorithms** for a scalable cybersecurity application and contributed backend functionality using Flask, REST APIs, and JavaScript.

Contingent Machine Learning Consultant July 2023 - March 2024
Meta
- Developed Python-based machine learning algorithms for intelligent task routing, applying Natural Language Processing (NLP) and **cosine similarity and re-ranking algorithms** to support task classification and assignment.
- Leveraged big data analytics to optimize task-triaging workflows, achieving a **78% reduction** in triage time and improving operational efficiency.
- Delivered high-priority machine learning projects within Agile methodologies, coordinating execution and priorities across cross-functional teams.
- Partnered with stakeholders to improve task-routing processes through data-driven analysis, algorithm development, and workflow optimization.

EDUCATION

Bachelor Of Science - Computer Science 05/2026
Virginia Tech, Blacksburg, VA, VA

PROJECTS

QEAgents | Agentic AI-Powered QA Automation | Ellucian Banner & Colleague
• Reduced test automation costs by 90% by transitioning QA workflows to an agentic pipeline spanning requirement analysis, test planning, test generation, test healing, and failure triage.
• Built custom local MCP integrations for Jira, Smartsheet, and SpiraTest, alongside Playwright MCP and GitHub tooling, to connect agents with enterprise systems and automation artifacts.
• Implemented specialized AI agents and human-in-the-loop checkpoints, with local execution, AWS Bedrock inference, Git-based artifact handoffs, and Jenkins CI/CD triggers.
DBWorkout | LLM-Powered SQL Learning Platform
• Developed an internal learning platform for Virginia Tech and the University of Virginia, combining an interactive SQL sandbox with an LLM-powered learning coach.
• Designed functionality for instructors to upload PDF materials and generate learning modules with knowledge checks and practical SQL exercises.
QualiHire | AI-Powered Resume & Skills Matching
• Built a resume skill-matching application that scores candidate and employee resumes against employer-submitted job requirements.
• Enabled recruiters and internal management to review candidate fit and support hiring decisions and internal resource alignment.

CERTIFICATIONS

- AccelQ Certification, AccelQ

- Azure Cloud Computing Fundamentals, Microsoft
- Stanford Data Science in Python, Coursera
- Machine Learning and Data Science in Python, Udemy